

Aryan Choudhary

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👤 ABOUT ME

Computer Science undergraduate who thrives on **shipping real products** that solve tangible problems for end users. From deploying live campus applications used daily to engineering high-performance distributed systems, I am deeply fascinated by **AI Engineering** and its potential to build intelligent, scalable solutions. Having leveraged large language models to streamline workflows and craft highly intuitive user experiences, I am eager to apply my software and AI engineering skills to the financial technology space to build innovative, quantitative, data-driven tools.

🎓 EDUCATION

Scaler School of Technology

UG in CSE

Bengaluru, India

Jul 2023 - May 2027

Birla Institute of Technology and Science, Pilani

B.S. in Computer Science

Pilani, India

Jul 2023 - May 2027

🧰 EXPERIENCE

TalkingLands

Software Engineer Intern

Bengaluru, India

Dec 2024 - Feb 2025

- Built a **React Native application** for real estate buyers, helping users evaluate locations through spatial mapping and reducing unnecessary field visits.
- Optimized application performance with a strong focus on **user experience**, reducing perceived waiting time and improving responsiveness.
- Designed and implemented dynamic product features and a custom newsfeed system using React and Strapi CMS.
- Developed a FastAPI proof-of-concept backend to validate workflows and external API integrations.

Dukan

AI SDE Intern

Bengaluru, India

Aug 2024 - Oct 2024

- Built an AI-powered hotel booking chatbot using OpenAI function calling, a **self-initiated project** that led to securing the internship.
- Utilized advanced AI engineering workflows with **Claude Pro (Team Plan)** to rapidly prototype, debug, and accelerate the development of complex backend features.
- Engineered a scalable **RAG-based retrieval pipeline** using vector embeddings and Wasabi S3, enabling business owners to query their documents in natural language.
- Designed and shipped the Bot9.ai landing page end-to-end, achieving a **perfect Google Lighthouse score of 100** across performance, accessibility, SEO, and best practices.

🔗 PROJECTS

FactorLab-Pro 🐍 — Python, CVXPY, SEC EDGAR, Plotly, Next.js

- Engineered a point-in-time US equity research pipeline combining SEC XBRL fundamentals with adjusted market data to construct value, quality, momentum, low-volatility, and investment factors.
- Implemented cross-sectional factor modeling with sector-relative winsorization, size/sector neutralization, rank information coefficients (IC), and factor-decile monotonicity tests.
- Developed constrained long-only and dollar-neutral portfolios using **CVXPY**, integrating position limits, sector neutrality constraints, turnover-based costs, and market-impact capacity estimates.
- Built a production-ready **Next.js** research terminal and automated pipeline featuring Parquet caching, Zod validation, and interactive **Plotly** tear sheets for exposure attribution.

Refection — *React Native, Node.js*

- Co-built a campus food pre-ordering platform that helps students avoid long canteen queues between classes.
- Implemented a scheduled pickup workflow allowing students to place orders in advance and collect meals at designated time slots.
- Currently **live on the Play Store with 100+ downloads** and a **4.4-star rating**, serving real users on campus daily.

KV Cache Server — *Python, asyncio, TCP, Docker*

- Built a high-performance **in-memory key-value database from scratch** using raw TCP sockets and asynchronous networking.
- Achieved nearly **12,000 requests per second with zero errors** during AWS-based load testing.
- Containerized the system with Docker for simplified deployment and scalability.

Distributed Live Polling System — *Redis, Python, Consistent Hashing*

- Built a distributed backend capable of processing **more than 10,000 votes per second** with **sub-5ms latency**.
- Reduced infrastructure load by **approximately 80%** through intelligent write batching and shard-aware request routing.

LEADERSHIP & ACTIVITIES

Chess Club, Scaler School of Technology

Bengaluru, India

President

2023 - 2024

- **Led club activities, organized tournaments**, and coordinated community events for students across batches.

SKILLS & TECHNOLOGIES

Languages: Python, Java, JavaScript, SQL, HTML/CSS

AI & LLM Engineering: Claude Pro, OpenAI Codex, Function Calling, RAG Pipelines, Vector Embeddings, Prompt Engineering

Finance & Quant: Factor Analysis, Backtesting (CVXPY), Time-Series Analysis, Portfolio Optimization

Frameworks & Backend: React, React Native, Next.js, Node.js, Express.js, FastAPI, Spring Boot

Databases & Infrastructure: PostgreSQL, MongoDB, Redis, Docker, AWS

Tools & Libraries: Git, Pandas, NumPy, OpenCV, Plotly, Matplotlib